GKE Lab: Persistent Volumes (PV and PVC)

In this lab you create a PersistentVolumeClaim (PVC), let GKE provision a PersistentVolume (PV) and a Persistent Disk for it automatically, write data from a Pod, and prove that the data survives when the Pod is deleted and recreated.

Prerequisites

- A running GKE cluster with kubectl connected. This guide uses mario-cluster in zone asia-south1-a from the previous lab.
- Cloud Shell or a terminal with gcloud and kubectl.

```
gcloud container clusters get-credentials mario-cluster
--zone=asia-south1-a
```

Objects in this lab

Object	Name	Purpose
StorageClass	standard-rwo	Built into GKE. Tells GKE to create a Balanced Persistent Disk.
PersistentVolumeClaim	data-pvc	Requests 10 GiB of ReadWriteOnce storage.
PersistentVolume	pvc-xxxx (auto-named)	Created by GKE and bound to data-pvc.
Pod	pv-demo	Runs nginx and mounts the volume at /usr/share/nginx/html.

Step 1: Check the available StorageClasses

```
kubectl get storageclass
```

You should see standard-rwo, usually marked (default). Note its VOLUMEBINDINGMODE is WaitForFirstConsumer, which matters in Step 2.

Step 2: Create the PVC

```
cat <<EOF > pvc.yaml
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: data-pvc
spec:
  storageClassName: standard-rwo
```

```
  accessModes:
  - ReadWriteOnce
  resources:
    requests:
      storage: 10Gi
EOF

kubectl apply -f pvc.yaml
kubectl get pvc data-pvc
```

Expected: STATUS is Pending. This is normal. With WaitForFirstConsumer, GKE waits until a Pod uses the claim, so it can create the disk in the same zone as that Pod.

Step 3: Create a Pod that uses the PVC

```
cat <<EOF > pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: pv-demo
spec:
  containers:
  - name: web
    image: nginx:1.27
    volumeMounts:
    - name: data
      mountPath: /usr/share/nginx/html
  volumes:
  - name: data
    persistentVolumeClaim:
      claimName: data-pvc
EOF

kubectl apply -f pod.yaml
kubectl wait --for=condition=Ready pod/pv-demo --timeout=180s
```

Step 4: Verify the PV and the disk were created

```
kubectl get pvc data-pvc
kubectl get pv
gcloud compute disks list --filter="name~^pvc-"
```

Expected: the PVC is now Bound, a PV named pvc-xxxx exists with CLAIM default/data-pvc and RECLAIM POLICY Delete, and a matching Persistent Disk appears in Compute Engine. You did not create the PV or the disk yourself; this is dynamic provisioning.

Step 5: Write data to the volume

```
kubectl exec pv-demo -- sh -c 'echo "Hello from a Persistent Volume"
```

```
> /usr/share/nginx/html/index.html'
kubectl exec pv-demo -- cat /usr/share/nginx/html/index.html
```

Step 6: Delete the Pod and prove the data persists

```
kubectl delete pod pv-demo
kubectl get pvc data-pvc
kubectl apply -f pod.yaml
kubectl wait --for=condition=Ready pod/pv-demo --timeout=180s
kubectl exec pv-demo -- cat /usr/share/nginx/html/index.html
```

Expected: the PVC stays Bound while the Pod is gone, and the new Pod prints "Hello from a Persistent Volume". The file lived on the Persistent Disk, not inside the container, so it survived.

Step 7 (optional): Expand the volume

standard-rwo allows volume expansion. Increase the claim to 20 GiB:

```
kubectl patch pvc data-pvc -p
'{"spec":{"resources":{"requests":{"storage":"20Gi"}}}}'
kubectl get pvc data-pvc
kubectl exec pv-demo -- df -h /usr/share/nginx/html
```

After a short while the CAPACITY column and df output show the larger size, without recreating the Pod.

Step 8: Clean up

```
kubectl delete pod pv-demo
kubectl delete pvc data-pvc
kubectl get pv
gcloud compute disks list --filter="name~^pvc-"
```

Because the reclaim policy is Delete, removing the PVC also deletes the PV and the Persistent Disk. Both lists should come back empty after a minute. If you want a disk to survive deletion of its claim, use a StorageClass with reclaimPolicy: Retain.

Key points

- A PVC is a request for storage; a PV is the storage that satisfies it.
- With a StorageClass, GKE creates the PV and Persistent Disk automatically when a Pod first uses the PVC.
- Data on a PV outlives the Pod. It is removed only when the PVC is deleted and the reclaim policy is Delete.
- ReadWriteOnce means one node can mount the volume at a time. For shared ReadWriteMany storage on GKE, use Filestore.